

WAN NUR SYAMILAH BINTI WAN ALI

PhD Candidate in Artificial Intelligence & Optimization
Artificial Intelligence | Data Mining | Computer Networks | Programming
Kuantan, Pahang | wansyamilah07@gmail.com | +601 3928 9006
ORCID: 0009-0000-0275-3944 | [Google Scholar Profile](#)

PROFESSIONAL SUMMARY

Researcher and educator with expertise in artificial intelligence, optimization, data mining, and computer networks. PhD candidate at UKM with a completed thesis on NP-hard incentive allocation optimization using World Bank datasets. Experienced in teaching, academic research, and applied machine learning across financial analytics, air quality modelling, and deep learning. Author of multiple Scopus- and WoS-indexed publications.

EDUCATION

PhD in Artificial Intelligence & Optimization

2021 – Present

Universiti Kebangsaan Malaysia (UKM)

- Research: Incentive allocation optimization for informal sector businesses (NP-hard combinatorial problem)
- Dataset: World Bank Indonesian Survey of Informal Sector Enterprises (ISES 2023)
- Thesis writing completed; pending final administrative clearance for submission

Master of Informatics (MSc)

2018 – 2020

Universiti Sains Malaysia (USM) | CGPA: 3.27

- Specialisation: Business Intelligence & Data Mining, Healthcare Informatics, ERP Systems, Financial Management
- Final project: Informatics research project (Grade A-)

Bachelor of Computer Science (Hons.) – Computer Networks

2021 – 2026

Universiti Teknologi MARA (UiTM) | CGPA: 3.37 | Flexible Learning Programme

- Completed whilst actively pursuing PhD research — pursued concurrently to strengthen applied CS and networking foundations
- Specialisation: Networking, IoT Systems, TCP/IP, Routing & Switching, Wireless Networks, Digital Forensics
- Key courses: Internet of Things (A), Network Programming, IP Routing, Network Operations

Bachelor of Science with Honours – Physics with Electronics

2013 – 2016

Universiti Malaysia Sabah (UMS)

- Covered: Digital Electronics, Microprocessors, Computational Physics, Instrumentation, Statistical Physics

INDUSTRY EXPERIENCE

IT Intern – System Developer

September – December 2025

Gagasan Consultant, Kuantan, Pahang (Civil & Structural Engineering Consultancy)

- Designed and developed a full-stack project database management system (MySQL, PHP, Apache/MAMP) to replace manual record-keeping
- Built a secure admin panel with CRUD operations, authentication, SQL injection prevention, XSS protection, and search/filter functionality across project type, year, and status
- Developed an analytics dashboard with visual charts (projects by year, type, and status)
- Implemented bilingual (English/Bahasa Malaysia) support throughout the system interface
- Designed and deployed a responsive company website from scratch using HTML, CSS, Figma, with cross-browser compatibility and mobile responsiveness

- Produced full technical documentation: requirements analysis, installation & deployment guide, maintenance guide, and test report

TEACHING EXPERIENCE

Primary School Teacher

2017 – 2018

Sekolah Rendah Islam Ibnu Abbas, Kuantan, Pahang

- Taught Science and Mathematics to Year 3, Year 5, and Year 6 pupils
- Delivered structured lesson plans across multiple subjects and year groups simultaneously

ACADEMIC & RESEARCH EXPERIENCE

Research Assistant (RA) – Data Mining

2020 – 2021

Universiti Kebangsaan Malaysia (UKM) | Supervisor: Prof. Dr. Zulaiha Ali Othman

- Project 1 — Household Overspending Model: Applied and compared five ML classifiers (J48, Naive Bayes, ANN, SVM, kNN) on Malaysian DOSM household expenditure data (4M records, 14,451 samples) to identify overspending patterns across B40/M40/T20 income groups
- Project 2 — Ozone Exceedance Prediction: Analysed 761,608 hourly air quality instances from 9 Klang Valley monitoring stations (DOE Malaysia, 2005–2016) using feature selection (Relief, InfoGain, GainRatio, OneR), classification (J48, REPTree, LMT, Random Forest, Random Tree), and Apriori association rules
- Conducted all experiments in WEKA; performed data preprocessing, discretization, imbalanced data handling, and model evaluation (Accuracy, MAE, RMSE, F-measure, Kappa, ROC)
- Contributed to published and manuscript-stage research outputs under Prof. Dr. Zulaiha Ali Othman

Graduate Research Assistant (GRA)

2021 – 2024

Universiti Kebangsaan Malaysia (UKM)

- Co-contributed to academic book writing on entrepreneurship development for B40 home-based women entrepreneurs
- Facilitated technical and academic workshops for faculty members and postgraduate students
- Conducted research data collection and analysis using Python (NumPy, Pandas, Matplotlib, SciPy — statistical testing: Mann-Whitney U, Kruskal-Wallis, Spearman ρ , Bonferroni; Keras, Scikit-learn)

RESEARCH & PUBLICATIONS

Journal Articles

- Wan Ali, WNS & Ali Othman, Z. (2026). Home-based business lifecycle for the development of B40 women entrepreneurs. *Journal of Small Business and Enterprise Development*, 33(2), 451–475.
- Yuliansyah, H. et al. (2026). Plastic Bottle Defect Detection Based on CNN with MobileNetV2 Architecture. *Franklin Open*, 100522.
- Othman, AMH., Ab Razak, MF., Firdaus, A., Ramli, S. & Wan Ali, WNSW. (2025). Trends in IoT Intrusion Detection: A Bibliometric Analysis of Deep Learning Approaches. *International Journal on Advanced Science, Engineering & Information Technology*.
- Yuliansyah, H. et al. (2023). Artificial intelligence in malnutrition research: a bibliometric analysis. *Bulletin of Social Informatics Theory and Application*, 7(1), 32–42.
- Ali, WNSW & Abdullah, NL. (2021). Climate change: Climate literacy and response among USM students. *Turkish Journal of Computer and Mathematics Education*, 12(3), 2205–2210.

Conference Papers

- Yuliansyah, H. et al. (2025). Deep Learning-Based Classification of Plastic Waste Using Resin Identification Codes. 1st International Conference on Artificial Intelligence Technology.
- Ali, WNS. (2020). Climate Change: Climate Literacy and Response Among USM Students. 2nd Applied Informatics International Conference (AilC2020), UPM. [Presented & Published]

TECHNICAL SKILLS

Programming	Python (NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, Pathlib, Argparse), PHP, HTML, CSS
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Databases	MySQL, SQL
Networking Tools	Cisco Packet Tracer, GNS3, Wireshark, VMware, Linux
DevOps / Cloud	Docker, Azure ML, MAMP (Apache/PHP/MySQL local server)
Data & Analytics	Excel, SPSS, SQL (intermediate), WEKA (J48, Random Forest, SVM, Apriori, feature selection), Grafana, Figma
Design & Dev Tools	VS Code, CodePen, phpMyAdmin
Research Areas	AI & Machine Learning, IoT Systems, Computer Networks, Combinatorial Optimization, Data Mining

TEACHING COMPETENCIES

Able to teach across the following areas based on academic background and research experience:

- Computer Networks & Networking Fundamentals
- Internet of Things (IoT)
- Programming Fundamentals (Python, PHP, HTML/CSS)
- Database Systems & Web Development
- Artificial Intelligence & Machine Learning
- Data Communications & Information Systems
- Research Methodology

LANGUAGES

Bahasa Malaysia	<i>Native</i>
English	<i>Proficient — academic writing, research, and technical documentation</i>